

**REAL-TIME TRAFFIC DETECTION AND SPEED ESTIMATION IN KENYA USING
YOLOv8**

BY:

NEEMA NDANU MUSOMBA

UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

SPRING 2025

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NEEMA NDANU MUSOMBA

**A Research Proposal Submitted to School of Science and Technology in Partial Fulfillment
of Requirements for the Award of Degree of Bachelor of Science in Data Science and
Analytics**

UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

SPRING 2025

STUDENT'S DECLARATION

I, the undersigned, declare that this research is my original work, except for cited quotations and references that has properly been acknowledged and referenced in accordance with the USIU-Africa requirements. Additionally, this research has not been submitted to any other college, institution, or any other university for examination, award of degree or publication or academic credit.

Signed:

Date:

Neema N Musomba (ID No 666457)

SUPERVISOR APPROVAL

This research proposal has been submitted for examination with my approval as the University Supervisor

Signed:

Date:

Dr. Gabriel Okello (ID No)

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ABSTRACT

Road Traffic Accidents (RTAs) are a major public health and economic issue in Kenya, with a fatality rate of 28.2 per 100,000 people and a 2.82% increase in deaths reported in 2024 (NTSA, 2024). Current traffic monitoring systems rely heavily on post-incident reports and static surveillance, which delays emergency response and limits real-time decision-making. This study addresses this challenge by developing a real-time accident detection and speed tracking system using the YOLOv8 object detection model and video data from the X platform (formerly Twitter).

The main objectives are to detect and classify road users (vehicles, motorcycles, pedestrians, and accidents), estimate vehicle speeds in real-time, and generate insights to support timely responses and policy decisions. The project uses a quantitative experimental approach, applying computer vision and machine learning techniques. Video clips shared online are processed using OpenCV, annotated for YOLOv8 training, and analyzed using a centroid-based tracking algorithm to estimate speed. The system is deployed through a Streamlit web app, enabling real-time visualization of detected objects and speed estimates.

System performance is evaluated using standard computer vision metrics: **mAP50** (mean Average Precision at 50% overlap), which measures object detection accuracy, and **MAE** (Mean Absolute Error), which reflects the difference between estimated and actual vehicle speeds. The model aims for $mAP50 > 0.95$ and $MAE < 5$ km/h. Ethical considerations, including privacy and consent, are addressed in line with Kenya's Data Protection Act (2019). This study offers a scalable, AI-driven framework for enhancing road safety, reducing emergency response times, and informing data-driven traffic policies in Kenya and similar low-resource settings.

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DEDICATION

This work is dedicated to all Kenyan citizens who have lost their lives or been affected by road traffic accidents. May this study contribute, in some small way, to the ongoing efforts to create safer roads and prevent future tragedies. I also dedicate this research to all road users and advocates working toward a safer transport system in Kenya, your resilience and commitment continue to inspire meaningful change.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADAS	Advanced Driver Assistance Systems
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
FPS	Frames Per Second
GMM	Gaussian Mixture Model
ID	Identification
IRB	Institutional Review Board
JPEG	Joint Photographic Experts Group (image format)
KES	Kenyan Shilling
MAE	Mean Absolute Error
mAP	Mean Average Precision
mAP50	Mean Average Precision at 50% Intersection over Union
NTSA	National Transport and Safety Authority
OpenCV	Open Source Computer Vision Library
PII	Personally Identifiable Information
RTA	Road Traffic Accident
SE Attention	Squeeze-and-Excitation Attention
SPD-Conv	Spatial Pyramid Dilated Convolution
YOLO	You Only Look Once
YOLOv8	Version 8 of You Only Look Once object detection model

CHAPTER 1: INTRODUCTION

1.1 Introduction

This chapter entails the problem's background of the problem, provide a clear knowledge of the statement of the problem, general objective of the study, research question, importance of the study, scope of the investigation, expected contribution of the project, and the chapter summary.

1.2 Background of the study

Road Traffic Accidents (RTAs) remain a significant public health and development concern globally, particularly in low- and middle-income countries where infrastructure and enforcement mechanisms often lag behind motorization growth. Globally, RTAs contribute to substantial economic losses, estimated at around 3% of a country's gross domestic product (GDP), due to costs related to health care, productivity loss, and infrastructure damage (Chand et al., 2021). The World Health Organization (WHO) reports that approximately 1.19 million people lose their lives annually due to traffic crashes, with millions more suffering non-fatal injuries—many of whom are left with long-term disabilities (WHO, 2023).

Sub-Saharan Africa bears a disproportionately high burden of road traffic deaths, with the region having the highest road traffic fatality rate in the world—approximately 19.4 deaths per 100,000 population (Taherpour, 2023). Kenya reflects this grim statistic. The National Transport and Safety Authority (NTSA) recorded 14,926 fatalities in 2021, resulting in a fatality rate of 28.2 per 100,000 people (NTSA, 2024). Between January and July 2024 alone, 2,407 fatalities were reported, representing a 2.82% increase compared to the same period in 2023. The majority of victims are young adults between the ages of 15 and 44 years, with a striking gender imbalance—85% of the victims are male (WHO, 2023).

Motorcycles and other vulnerable road users such as pedestrians and cyclists account for a large proportion of the deaths and injuries. Specifically, 28% of fatalities involve riders of two- or three-wheeled vehicles, 26% are occupants of four-wheeled vehicles, 24% are pedestrians, and 2% are cyclists (NTSA, 2024). Muguro et al. (2020) reported a 26% increase in fatalities and a

46.5% increase in injuries from January 2015 to January 2020, with vulnerable road users seeing the most dramatic rise.

In response to this alarming trend, global and national initiatives have been introduced to promote road safety. The United Nations launched the Global Plan for the Decade of Action for Road Safety (2011–2020), aiming to reduce road traffic deaths and injuries by 50% (Chang et al., 2020). Road safety also features under Target 3.6 of the United Nations Sustainable Development Goals (SDGs), which seeks to halve global deaths and injuries from RTAs by 2030.

Despite these interventions, RTAs in Kenya remain persistent and often under-monitored in real-time, leading to delayed emergency responses and missed opportunities for proactive intervention. Traditional monitoring systems rely heavily on post-accident reports, eyewitness accounts, and static surveillance infrastructure, which often lack precision and scalability.

Advancements in Artificial Intelligence (AI) and Computer Vision, particularly real-time object detection models like YOLO (You Only Look Once), offer promising solutions for proactive monitoring and intervention. YOLO is a state-of-the-art, real-time object detection system that can identify and classify objects, such as vehicles and pedestrians, and track their speed and movement in video feeds with high accuracy and speed. Leveraging such technologies in Kenya could revolutionize how traffic surveillance and accident monitoring are conducted, enabling faster emergency responses, enhancing law enforcement capabilities, and potentially saving lives.

Therefore, the study seeks to explore the application of real-time object detection and speed tracking using YOLOv8 to improve road safety in Kenya. It aims to develop a data-driven system capable of identifying and analyzing traffic accidents in real-time, thereby providing actionable insights for intervention and policy development.

1.3 Problem of the Statement

Kenya is among the African countries with the highest rates of road traffic accidents, largely due to reckless driving, poor enforcement of traffic laws, inadequate infrastructure, and the absence of timely accident response mechanisms. Despite ongoing efforts by stakeholders such as the

National Transport and Safety Authority (NTSA), road traffic accident (RTA) fatalities and injuries continue to rise, particularly among vulnerable road users like motorcyclists and pedestrians (NTSA, 2024).

Existing research and policy interventions have primarily focused on analyzing historical accident data, identifying long-term trends, and enforcing traffic regulations (Muguro et al., 2020; Zheng et al., 2020). However, these approaches lack the immediacy and contextual awareness needed to prevent or mitigate accidents in real time. There is a critical gap in the ability to automatically detect accidents as they occur, assess contributing factors such as vehicle speed and object movement, and respond promptly.

The absence of real-time surveillance and AI-powered analytical tools significantly hinders rapid decision-making and emergency response. Additionally, most available models are designed for structured, high-income environments and fail to capture the complexity of Kenya's mixed traffic systems, including informal transport and chaotic urban congestion.

Unlike prior studies focused on post-accident analysis or simulations, this project introduces a pioneering approach to real-time accident detection and speed tracking using the YOLOv8 deep learning model in a low-resource Kenyan setting. By leveraging publicly available video footage from the X platform (formerly Twitter), the system enables real-time identification of RTAs, classification of involved road users, and estimation of vehicle speed without the need for fixed infrastructure.

This AI-driven solution addresses a crucial technological gap by offering a scalable, locally adaptable tool that enhances surveillance, accelerates emergency response, and supports data-driven policymaking to reduce road traffic accidents in Kenya.

1.4 General Objective

The primary goal of this study is to develop a real-time road traffic accident detection and speed tracking system using the YOLOv8 object detection algorithm. The system aims to enhance road safety in Kenya by identifying accidents as they occur and providing critical data on vehicle speed and object movement to support timely emergency response and data-driven policy decisions.

1.4.1 Specific Objective

1. To develop a YOLOv8-based system capable of detecting and classifying key road users, such as vehicles, motorcycles, and pedestrians, from real-time video footage.
2. To implement a speed estimation algorithm that calculates vehicle velocities based on object tracking and assesses risk factors leading up to potential collisions.
3. To integrate object detection and speed tracking into a unified platform that provides real-time alerts and visualizations to aid rapid response and stakeholder decision-making.

1.5 Research Questions

- How can real-time video surveillance and object detection techniques be used to identify road traffic accidents as they occur in Kenya?
- How effective is the YOLO-based system in detecting and classifying road users and estimating vehicle speeds in accident scenarios?
- How can data collected from real-time object detection and speed tracking inform timely interventions and long-term road safety strategies in Kenya?

1.5.1 Significance of the study

The study on road traffic accidents (RTAs) in Kenya, has significant implications and holds great promise for various groups:

1.5.2 Government and policy Makers

Government agencies such as the Ministry of Transport and NTSA will benefit from real-time, data-driven insights that can inform infrastructure development, strengthen road safety policies, and guide the deployment of intelligent traffic monitoring systems.

1.5.3 Law Enforcement and Traffic Management Authorities

Police departments and traffic management units can use the proposed system to monitor traffic in real-time, enforce speed regulations, and respond more quickly to accidents. This can reduce emergency response time and improve overall road discipline.

1.5.4 HealthCare and Emergency Response Services

The system supports quicker mobilization of medical and emergency teams by providing immediate accident detection and location data. Hospitals and ambulance services can use this information for better triage, preparedness, and resource allocation.

1.5.5 Public and Commuters

Road users—drivers, passengers, pedestrians—will benefit from increased safety through better-informed travel decisions, awareness of high-risk areas, and faster accident response, thereby reducing injury severity and fatalities.

1.5.6 Economic Stability and Workforce protection

By reducing accident-related fatalities and injuries, the system protects workforce productivity and minimizes healthcare costs, property damage, and other economic losses associated with RTAs.

1.6 Scope of the study

The study aims to assess the potential of real-time video surveillance to detect accidents, track vehicle speeds, and analyze object movements, all of which are critical for improving road safety in Kenya. Geographically, the study focuses on major urban and high-traffic areas in Kenya, with particular attention to road networks where road traffic accidents (RTAs) are most prevalent. Data is collected from real-time video feeds, leveraging machine learning models to analyze traffic patterns, detect accidents, and estimate vehicle speeds.

Specifically, the study examines the functionality and accuracy of the YOLO model in detecting various road users, such as vehicles, motorcycles, and pedestrians, while tracking their speeds during potential accidents. Additionally, the study explores the integration of these real-time data points into a monitoring system capable of providing immediate alerts to stakeholders.

The research will be conducted from January to August 2025, with findings aimed at informing future interventions and technological advancements in road traffic safety.

1.7 Definition of Terms

1.8.1 Accidents

Berhanu, Y., Alemayehu, E., and Schröder, D. (2023) define an accident as a widely accepted term describing an incident involving one or more vehicles in a collision, leading to property damage, injury, or loss of life.

1.8 Expected Contributions of Study

- Demonstrating the application of real-time video analysis and object detection models in identifying and monitoring road traffic accidents (RTAs) in Kenya.
- Improving the understanding of traffic dynamics by analyzing vehicle movements, collision events, and speed patterns in high-traffic urban areas.
- Providing technical insights for policymakers, urban planners, and traffic enforcement agencies on how AI-driven tools can be integrated into existing traffic surveillance infrastructure to enhance road safety.
- Validating the effectiveness of machine learning techniques, particularly YOLO, in detecting various road users (e.g., vehicles, motorcycles, pedestrians) and estimating vehicle speeds during potential accident scenarios.
- Supporting the development of automated, real-time alert systems that can notify relevant authorities of detected road incidents to facilitate faster response times and preventive measures.
- Laying the foundation for future smart traffic monitoring systems that incorporate real-time object detection and speed tracking for improved decision-making and policy development.

1.9 Chapter Summary

This chapter provides an overview of the research problem, emphasizing the significance of road traffic accidents (RTAs) in Kenya and the need for real-time technological solutions to improve road safety. It introduces the study by highlighting the impact of RTAs on public health, mobility, and economic productivity, and the potential role of computer vision and machine learning in mitigating these effects. The chapter outlines the problem statement, research objectives, key questions, and justification for focusing on real-time object detection and speed tracking.

It also defines the scope of the study, centering on the application of the YOLO model for detecting road users, monitoring vehicle speeds, and identifying potential accidents using real-time video feeds. The chapter concludes with the expected contributions, emphasizing the study's role in advancing intelligent traffic monitoring systems and providing actionable insights for stakeholders. The subsequent chapters will cover the literature review, research methodology, system design, data analysis, and study findings.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of previous research related to real-time road traffic monitoring, focusing on accident detection, object tracking, and vehicle speed estimation using computer vision and machine learning techniques. The review explores key technologies and methodologies used in traffic accident detection systems, including image and video processing, object detection models, speed estimation algorithms, and accident recognition techniques. The chapter is organized into four sections: The General Literature Review, which presents an overview of existing studies; the Empirical Review, which examines specific research and methodologies; the Research Gaps section, which highlights unexplored areas; and the Chapter Summary, which consolidates key findings.

2.2 General Literature Review

2.2.1 Introduction of Detection models in Computer Vision

Object detection is a foundational and impactful component of computer vision. It plays a critical role in powering real-time applications such as traffic monitoring, surveillance systems, autonomous vehicles, and smart cities. By enabling machines to identify and locate objects within images or video streams, object detection supports the automatic recognition of vehicles, pedestrians, and other relevant entities in dynamic environments.

Chung et al. (2020) emphasize that the rapid advancement of deep learning has significantly driven the evolution of object detection technologies. At the heart of this progress are Convolutional Neural Networks (CNNs), which excel in extracting hierarchical features from visual data. These features make CNNs particularly effective at recognizing patterns, shapes, and object boundaries.

According to Wu et al. (as cited in Khazukov et al., 2020), object detection models based on neural networks can generally be classified into two categories: single-stage and two-stage models.

Single-stage models—such as YOLO (You Only Look Once), SSD (Single Shot Detector), and RetinaNet—perform localization and classification simultaneously in a single network pass. These models are optimized for speed, making them ideal for real-time applications, although they may compromise accuracy, especially when detecting small or overlapping objects.

Two-stage models, including R-CNN, Fast R-CNN, and Faster R-CNN, operate in two phases: they first generate region proposals likely to contain objects, and then classify and refine these proposals. While more accurate, these models are typically slower and computationally intensive.

The development of R-CNN in 2014 marked a turning point, introducing a structured approach to object detection by focusing on regions of interest (RoIs) within images (Chung et al., 2020). Subsequent innovations, such as Fast R-CNN and Faster R-CNN, further optimized detection speed and computational efficiency while maintaining high accuracy, enhancing their suitability for practical applications.

Further advancements have combined detection models with tracking algorithms to monitor object movement over time—a critical feature in traffic surveillance. For instance, the Mask R-CNN model introduced by He et al. (2017) expanded on Faster R-CNN by adding instance segmentation, allowing not only bounding box detection and classification but also pixel-level contour identification. This integration provides a deeper semantic understanding of visual scenes.

Despite significant progress, object detection systems still face challenges in real-world environments. Adverse conditions—such as low visibility, nighttime scenes, poor weather (e.g., rain, fog, snow), and occlusions—can hinder performance. These challenges highlight the ongoing need for models that are robust, adaptive, and capable of generalizing across diverse operational conditions.

2.2.2 Speed Tracking

Speed tracking involves monitoring and measuring the velocity of moving objects—particularly vehicles—over time. This capability is vital in traffic surveillance and accident analysis, where reconstructing vehicle speed prior to collisions supports fault determination and emergency

planning. Beyond reactive measures, accurate speed tracking also enables proactive road safety by identifying high-risk behaviors and hazardous locations.

To facilitate speed tracking, various algorithms have been developed. Traditional approaches such as the Gaussian Mixture Model (GMM) remain widely used for motion detection. As Lin et al. (2021) explain, GMM models the background of a video and detects changes over time, allowing it to isolate moving objects from static scenes. This separation is crucial for estimating vehicle speed in surveillance systems.

In recent years, deep learning models like YOLO have gained prominence due to their ability to simultaneously detect and classify objects in a single network pass. YOLO's fast processing makes it especially well-suited for real-time speed estimation. By tracking vehicle positions across consecutive frames, systems can calculate displacement over time, which allows for reliable speed computation—an essential metric in traffic management and safety systems.

2.2.3 Advancement of the models

Over the past decade, object detection has seen remarkable advancements, particularly through the evolution of single-stage models like the YOLO family. These models revolutionized real-time detection by removing the need for complex region proposal steps typical of earlier multi-stage detectors such as the R-CNN family.

As noted by Mutabarura, Muchuka, and Segeera (2025), the original YOLO algorithm introduced a transformative approach by predicting bounding boxes and class probabilities directly from full images in a single forward pass. However, early versions of YOLO struggled with precise localization, especially in scenes with small or densely packed objects. This limitation arose from its grid-based prediction mechanism, which limited accuracy.

To address these issues, the YOLO framework has undergone significant refinement. Successive versions—YOLOv2 through YOLOv10—have introduced architectural improvements aimed at boosting both speed and accuracy.

For example: YOLOv2 incorporated batch normalization, high-resolution classifiers, and multi-scale training to improve localization and speed. YOLOv3, built on the deeper Darknet-53

network, added residual connections and multi-scale predictions for better feature extraction. YOLOv4, as detailed by Li et al. (2023), introduced mosaic augmentation, self-adversarial training, and Cross-Stage Partial connections (CSPNet) to optimize performance in real-world settings. YOLOv5, discussed by Zhang et al. (2022), enhanced efficiency with automatic anchor box calculation, focus layers, and improved data augmentation techniques.

Subsequent iterations—YOLOv6 through YOLOv10—have continued this trajectory by integrating transformer-based architectures, improved backbone networks, and more advanced feature pyramid structures.

Nepal and Eslamiat (2022) highlight that post-2020 YOLO versions (especially YOLOv5 and beyond) are highly flexible and capable of detecting a diverse range of traffic participants, including vehicles, pedestrians, and cyclists. This makes them invaluable in applications such as autonomous driving, intelligent transportation systems, and traffic management—domains where both speed and precision are paramount.

These models have been trained and evaluated on large-scale public datasets, including PASCAL VOC, COCO, and ImageNet, which offer diverse object classes and challenging real-world conditions (Zendel, 2022). This has allowed YOLO-based systems to generalize effectively across varied environments.

In summary, the evolution of YOLO and other detection models reflects a broader trend in computer vision: toward faster, more accurate, and more adaptable systems. These models are increasingly central to a wide array of applications—from robotics and surveillance to environmental monitoring and healthcare—underscoring their importance in today's AI-driven world.

2.3 Empirical Literature Review

Empirical studies have demonstrated the efficacy of computer vision and machine learning in real-time road traffic accident (RTA) detection and speed tracking, with YOLO-based systems showing particular promise.

Lin et al. (2021) designed a real-time traffic monitoring system based on a virtual detection zone using Gaussian Mixture Models (GMM) and YOLOv4. The system improved vehicle counting and classification efficiency and estimated speed based on the distance and time traveled by a vehicle. It achieved classification accuracies of 99.1% (daytime), 98.6% (nighttime), and 98% (rainy conditions), with an average absolute percentage error of 7.6% in speed estimation. A key limitation, however, was the assumption that vehicles remain within predefined detection zones—an approach less suitable for dense urban environments like Nairobi, where vehicles frequently change lanes or directions. This is relevant to Kenya because the model's ability to function in rainy conditions aligns with local weather patterns. However, its dependence on fixed zones may not be viable in Nairobi's informal and unpredictable traffic system, where matatus and motorcycles often deviate from designated lanes.

Kumar and Pradhan (2022) used computer vision techniques, specifically OpenCV and dlib, to track and monitor vehicle speeds. By calculating displacement between video frames and measuring time intervals, the system accurately recorded vehicle motion and velocity. This is pertinent to Kenya, where precise speed tracking can support enforcement of traffic laws, especially amid a 3.1% rise in RTA fatalities in 2024 (Muguro et al., 2023). Nevertheless, the system's inability to handle occlusions and its limited robustness across diverse vehicle types—such as the ubiquitous motorcycles—poses challenges for application in Kenya's traffic landscape.

Murthy et al. (2022) introduced an object detection method using the YOLOv5 algorithm for Advanced Driver Assistance Systems (ADAS). Their approach outperformed YOLOv3 and YOLOv4 in terms of detection speed and accuracy, achieving 95% detection accuracy and superior frame rates, as measured by mean average precision (mAP) and frames per second (FPS). Deployed via a mobile application named Object Detect, the system offered users real-time alerts through a simple interface. This approach is relevant to Kenya, where mobile-based solutions could empower drivers with instant safety notifications. However, because the system was tested in controlled environments, it may not generalize well to chaotic urban intersections and pedestrian-heavy zones commonly seen in cities like Nairobi.

Huang (2020) proposed a two-stream convolutional neural network combining a spatial stream for object detection and a temporal stream that incorporates motion features for multi-object tracking. This design significantly improved near-accident detection accuracy by using both appearance and movement cues. The model is relevant to Kenya, as early identification of near-accidents could help prevent fatal crashes, especially in urban centers. However, the system's high computational demands may hinder its real-time use in Kenya, where access to powerful hardware is limited.

In another study, Rahman (2020) focused on detecting wrong-way driving using a YOLO-based model combined with centroid tracking. The system successfully identified and tracked vehicles from surveillance footage, making it useful for detecting anomalies in traffic flow. This is applicable to Kenya, where wrong-way driving is common on highways such as the Nairobi–Mombasa Road. However, ID-switching—caused by overlapping centroids in crowded scenes—reduces tracking reliability, particularly in Kenya's congested roads where multiple road users frequently intersect.

Shaqib (2024) proposed a cost-effective vehicle detection and speed tracking model using YOLOv8. The system demonstrated strong accuracy and efficiency, with a Mean Absolute Error (MAE) of 3.5 and a Root Mean Square Error (RMSE) of 4.22. Validated against actual speedometer readings, the model is promising for resource-constrained settings like Kenya. Still, the author highlighted the need for further testing in diverse traffic conditions to ensure adaptability across varying environments.

to overcome the challenge of detecting small and distant objects—common in existing RTA detection systems—Chen (2024) developed YOLOSED, an enhanced version of YOLOv8. The model integrates SPD-Conv and SE attention blocks to improve feature extraction and accuracy. Trained on a large dataset from Chinese urban surveillance systems, it performed well in detecting accidents. While promising, its applicability to Kenya remains uncertain due to untested performance in informal road networks and rural traffic settings, where conditions differ significantly from those in Chinese cities.

These studies highlight YOLO's strengths in real-time object detection and speed tracking, achieving high accuracy and low estimation errors across multiple implementations. However,

many of these models were developed and tested in structured environments with consistent traffic flow and access to advanced hardware. Their reliance on controlled datasets, predefined detection zones, or high-performance computing limits their transferability to low- and middle-income countries like Kenya. This is particularly concerning given Kenya's mixed traffic composition—including pedestrians, motorcycles, and informal public transport—that operates in unpredictable and often congested settings. Moreover, most existing solutions lack contextual adaptation or validation on Kenyan data, raising concerns about model robustness and generalization. These limitations underscore the need for a locally tailored, computationally efficient system.

This study addresses that gap by training and testing a YOLOv8-based detection and speed tracking system on real-world, user-generated Kenyan video data sourced from the X platform, offering a novel and scalable solution suitable for deployment in resource-constrained environments.

2.4 Research Gaps

The literature reviewed in Sections 2.2 and 2.3 highlights significant advancements in real-time RTA detection and speed tracking through computer vision and machine learning, particularly YOLO-based systems. These methodologies have improved accident detection, road user classification, and speed estimation, enabling data-driven traffic safety management. However, critical research gaps remain, particularly in addressing Kenya's unique road safety challenges, as outlined in Chapter 1.

2.4.1 Limited Contextualization for Kenya

The application of real-time road traffic accident (RTA) detection and speed tracking systems, particularly those using YOLO, has been predominantly explored in developed or urbanized regions such as China, Bangladesh, and the United States, leaving a significant research gap in low- and middle-income countries (LMICs) like Kenya. Kenya's traffic landscape, characterized by a high prevalence of motorcycles (30% of road users), substantial pedestrian risks (24% of fatalities), and a mix of formal and informal transport systems like matatus, presents unique challenges that generalized models fail to address (NTSA, 2025). Unlike the standardized traffic compositions and robust infrastructure in regions studied by Shaqib (2024) and Chen (2024),

Kenya's roads feature poor maintenance, inconsistent law enforcement, and diverse environmental conditions, such as rainy seasons, which demand context-specific solutions.

For instance, Shaqib (2024) demonstrated YOLOv8's effectiveness in Bangladesh, achieving a Mean Absolute Error (MAE) of 3.5 for speed estimation, while Chen (2024) introduced YOLOSED for accident detection in Chinese cities, enhancing small-object detection. However, these studies focus on urban settings with relatively uniform vehicle types, neglecting Kenya's rural roads and informal transport dynamics. Muguro et al. (2023) highlight Kenya's rising RTA trends, with a 3.1% fatality increase in 2024, yet tailored applications of computer vision remain scarce. The lack of region-specific datasets and models limits the applicability of existing research to Kenya, where socio-economic factors, such as low traffic regulation awareness, further complicate traffic patterns. Given that 93% of global RTA fatalities occur in LMICs (Sundet, 2023), developing YOLO-based systems that account for Kenya's unique traffic composition and infrastructure is critical for effective real-time surveillance and safety interventions.

2.4.2 Data Quality and Computational Constraints

The reliability of YOLO-based systems for real-time RTA detection and speed tracking depends on the quality, consistency, and availability of video footage, yet many studies face challenges due to the scarcity of structured, high-quality datasets in LMICs like Kenya. Unlike developed regions with extensive traffic camera networks, Kenya lacks centralized video surveillance, making traditional data collection difficult (Wanjiku, 2023). However, the widespread use of social media, particularly X, where Kenyans actively share real-time traffic incident videos, offers a promising solution. These user-generated videos capture Kenya's diverse traffic scenarios, including urban congestion and rural road dynamics, essential for training models to detect accidents and track speeds in a context with high motorcycle (30%) and pedestrian (24%) involvement (NTSA, 2025).

Despite this potential, studies like Huang (2023) and Shaqib (2024) rely on curated, high-quality video feeds from controlled settings, achieving accuracies up to 99.1% and speed estimation errors as low as 3.5 MAE. These datasets often lack the variability of X footage, which may include inconsistent resolutions, frame rates, or partial coverage due to user-uploaded content.

Preprocessing X videos to standardize quality and ensure comprehensive incident representation, especially in rural or low-visibility conditions, remains underexplored. Additionally, computational constraints pose significant barriers in Kenya's resource-limited settings. Real-time analysis using advanced YOLO models, such as YOLOv8 or YOLOSED, demands substantial processing power, which is often unavailable due to limited access to high-performance hardware or cloud computing (Chen, 2024). Optimizing YOLO models for computational efficiency and enhancing X video preprocessing techniques are essential to overcome dataset scarcity and enable scalable, reliable RTA monitoring in Kenya.

2.4.3 Challenges in Handling Diverse Traffic Scenarios

YOLO-based systems excel in controlled environments but are often trained on datasets from low-traffic or predictable settings, such as suburban roads or structured urban areas, which fail to capture the chaotic, high-density traffic scenarios prevalent in Kenya. These simplified datasets limit YOLO's ability to demonstrate its full potential in handling complex interactions, such as occlusions, rapid vehicle movements, and low-visibility conditions common in Kenya's urban and rural roads. Kenya's traffic, with motorcycles weaving through congestion, matatus navigating crowded streets, and pedestrians crossing unpredictably, requires models robust enough to process dynamic, multifaceted scenarios (NTSA, 2025).

For example, Lin et al. (2023) achieved 99.1% classification accuracy using YOLOv4, but their dataset from sparse traffic settings did not reflect Kenya's urban intersections, where overlapping vehicles cause occlusions. Similarly, Rahman (2024) noted centroid tracking errors in moderately dense traffic, a challenge amplified in Kenya's overcrowded cities like Nairobi. Chen (2024) and Shaqib (2024) used datasets from structured environments, missing the informal transport and rural road dynamics critical to Kenya. The X platform, with its real-time, user-generated videos of Kenya's dense traffic, offers a unique opportunity to train YOLO models on high-traffic scenarios, yet such datasets are rarely utilized. Computational demands for processing these complex videos further complicate deployment in Kenya, where infrastructure limitations restrict access to advanced computing resources. Developing YOLO models optimized for diverse traffic and computational efficiency is crucial to unlocking their full potential for real-time RTA detection and speed tracking in Kenya.

2.5 Chapter Summary

This chapter reviewed the extensive body of research on real-time road traffic accident (RTA) detection and speed tracking, emphasizing the role of computer vision and machine learning, particularly YOLO-based systems, in enhancing road safety. It explored general literature, empirical studies, and critical research gaps to provide a comprehensive understanding of current methodologies and their applicability to Kenya's unique traffic challenges.

Section 2.2 presented a general literature review, tracing the progression of object detection models from two-stage approaches, such as R-CNN and Faster R-CNN, to single-stage models like YOLO, which prioritize speed for real-time applications (Chung et al., 2020; Mutabarura et al., 2025). The review highlighted YOLO's advancements, from its initial grid-based predictions to modern iterations (YOLOv2–YOLOv10) incorporating batch normalization, multi-scale predictions, and transformer architectures, achieving high accuracy and flexibility (Nepal & Eslamiat, 2022; Li et al., 2023). It also explored speed tracking techniques, from Gaussian Mixture Models (GMM) to YOLO-based frame-by-frame displacement analysis, underscoring their importance in traffic management (Lin et al., 2021). The section emphasized challenges in real-world conditions, such as occlusions and low visibility, which limit model performance in diverse settings like Kenya.

Section 2.3 provided an empirical review of methodologies used in real-time RTA detection and speed tracking. Studies like Lin et al. (2023) and Shaqib (2024) demonstrated YOLO's efficacy, achieving classification accuracies up to 99.1% and speed estimation errors as low as 3.5 MAE, while innovations like YOLOSED enhanced small-object detection (Chen, 2024). Other approaches, such as Huang's (2023) two-stream network and Rahman's (2024) centroid tracking, improved detection and tracking capabilities. However, these studies often relied on controlled environments, faced computational complexity, and lacked testing in Kenya's mixed traffic, limiting their applicability to local conditions with high motorcycle (30%) and pedestrian (24%) involvement (NTSA, 2025).

Section 2.4 identified key research gaps, including limited contextualization for Kenya, where unique traffic compositions and poor infrastructure demand tailored models; insufficient use of diverse, high-traffic datasets, which restricts YOLO's potential; and data quality and

computational constraints, particularly in leveraging X video footage in resource-limited settings. These gaps highlight the reliance on low-traffic datasets, the scarcity of Kenya-specific research, and the challenges of processing user-generated videos, underscoring the need for innovation in real-time traffic surveillance.

Overall, this chapter established the need for a novel approach that leverages X-derived video datasets to capture Kenya's complex traffic dynamics, optimizes YOLO-based systems for diverse scenarios, and addresses computational constraints to enhance road safety interventions. The insights from this literature review will inform the research methodology and system design in subsequent chapters, paving the way for a context-aware, real-time RTA detection and speed tracking framework tailored to Kenya's needs.

CHAPTER 3.0: METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology employed to achieve the study's objectives, focusing on developing a real-time road traffic accident (RTA) detection and speed tracking system in Kenya. The methodology leverages computer vision techniques, specifically object detection and speed estimation, applied to video footage, to monitor traffic incidents and enhance road safety. The chapter is structured to cover research design, data collection, analysis methods, system development, validation, and ethical considerations, providing a robust framework for addressing the research gaps identified in Chapter 2.

3.2 Research Design

The research design serves as the overarching strategy to address the study's research questions, as outlined in Chapter 1, focusing on developing a real-time road traffic accident (RTA) detection and speed tracking system tailored to Kenya's diverse traffic environment. This study adopts a quantitative experimental methodology, leveraging computer vision and machine learning to design, implement, and evaluate a YOLOv8-based system for detecting vehicles, pedestrians, motorcycles, and accidents, and estimating vehicle speeds. The design emphasizes applied research to deliver a practical tool for stakeholders, such as the National Transport and Safety Authority (NTSA), to enhance traffic surveillance and road safety interventions.

The research design is structured around two interconnected components:

- **Case Study:** Kenya provides a real-world testbed for the YOLOv8 system in both urban and rural areas, where motorcycles and pedestrians are heavily involved in RTAs (NTSA, 2025). The study explores how real-time object detection and speed tracking can enhance traffic monitoring.
- **Prototype Development:** A Streamlit-based prototype processes video from the X platform to detect objects and track speed in real time, enabling data-driven traffic management decisions.

The quantitative experimental methodology dominates, utilizing numerical data extracted from derived video frames, including bounding box coordinates, object class probabilities, and

frame-to-frame displacements, to achieve precise detection and speed estimation. The YOLOv8 model, optimized for real-time performance, is fine-tuned on a custom dataset to address the research gaps identified in Chapter 2: limited contextualization for Kenya, data quality challenges with user-generated footage, and the need for models robust to diverse traffic scenarios. Controlled experiments, such as model training with defined parameters and validation using metrics like mean Average Precision (mAP) and Mean Absolute Error (MAE), ensure rigorous evaluation.

3.3 Design/Experiment Methodology

This section details the experimental process, emphasizing the technical workflow, models, and prototypes used to develop the RTA detection and speed tracking system. It outlines the step-by-step pipeline for data collection, preprocessing, model training, speed estimation, system deployment, and evaluation.

3.3.1 Models

YOLOv8

YOLOv8 is a state-of-the-art, single-stage object detection model optimized for speed and accuracy, ideal for real-time applications. It predicts bounding boxes and class probabilities for objects (e.g., cars, pedestrians, motorcycles, accidents) in a single forward pass, using a convolutional neural network (CNN) with a Darknet backbone. The loss function combines bounding box regression, classification, and distribution focal loss:

$$L = \lambda_{box} \times L_{box} + \lambda_{cls} \times L_{cls} + \lambda_{dfl} \times L_{dfl}$$

where:

- L_{box} : Bounding box regression loss (e.g., CIoU loss).
- L_{cls} : Classification loss (binary cross-entropy).
- L_{dfl} : Distribution focal loss for box refinement.
- $\lambda_{box}, \lambda_{cls}, \lambda_{dfl}$: Weighting coefficients.

For this project, YOLOv8 is fine-tuned on a custom dataset from X videos to detect four classes: Car, Pedestrian, Motorcycle, and Accident. The model's architecture includes mosaic augmentation, multi-scale predictions, and attention mechanisms to handle Kenya's diverse traffic scenarios.

Speed Tracking Algorithm

Speed tracking is implemented by calculating vehicle displacement across consecutive video frames. Using YOLOv8's bounding box coordinates, the centroid of each detected vehicle is tracked over time. The speed is computed as:

$$v = \frac{\Delta d}{\Delta t} \times s$$

where:

- Δd : Pixel displacement between centroids in consecutive frames.
- Δt : Time interval between frames (e.g., 1/FPS seconds).
- s : Scaling factor (meters per pixel), calibrated based on road dimensions.

The algorithm accounts for camera perspective and frame rate variability in X footage, ensuring accurate speed estimates in real-time.

The experimental process converts Twitter data into actionable road traffic accident (RTA) insights through a structured workflow, as shown in Figure 1: Process Map for RTA Analysis. This map outlines the progression from data collection to prototype deployment, guiding the five key phases. The following pipeline visually represents this workflow.

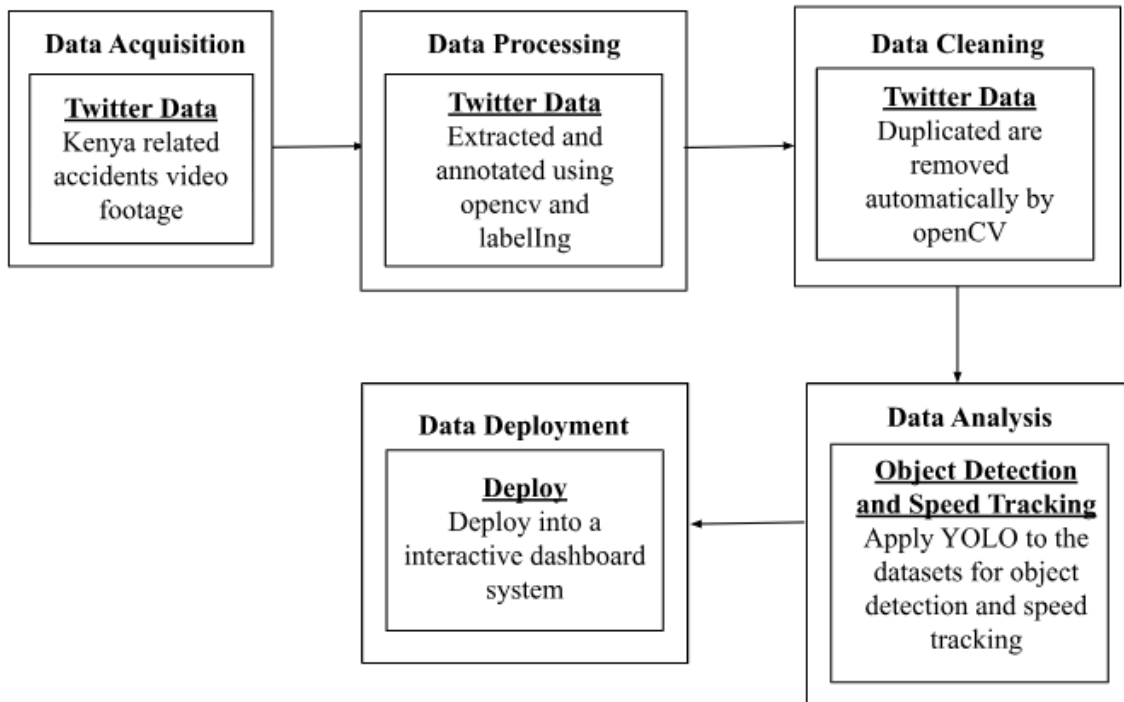


Figure 1: Process Map for RTA Analysis

3.4 Design/Experiment Procedures

3.4.1 Data Acquisition

The dataset will comprise video footage of Kenyan road traffic sourced exclusively from the X platform, leveraging its popularity among Kenyans for sharing real-time traffic incidents (NTSA, 2025). Videos posted between 2023 and 2025 will be collected from accounts such as @Ma3Route using keywords like “accident,” “crash,” and “traffic.” An online Twitter video downloader will be used to download the MP4 videos, which will then be saved locally. No specific timeframe will be defined to ensure a broad range of scenarios (urban, rural, day, night). To create a cohesive dataset, the downloaded videos will be compiled into a single composite video using FFmpeg, ensuring smooth transitions while preserving original content. Metadata such as video IDs and user-reported locations (e.g., “Nairobi”) will be manually recorded in a local text file, without relying on precise timestamps.

3.4.2 Data Processing

The composite video will be processed to extract frames for annotation and model training. Using OpenCV, every nth frame will be extracted, resized for optimal quality and efficiency, and saved as JPEG images. Labelling will be used to manually annotate frames in YOLO format, labeling objects as Car (1), Pedestrian (2), Motorcycle (3), or Accident (4), with bounding box coordinates and class IDs saved as text files. The dataset will be organized into training (80%) and validation (20%) sets within a folder, which will later be uploaded for model training. A data.yaml file will define the dataset structure, specifying paths, number of classes, and class names.

3.4.3 Data Cleaning

To ensure data quality, frames with low resolution, excessive noise, or irrelevant content such as non-traffic scenes will be filtered out using OpenCV's image processing functions. Duplicate frames—especially those at the transition points between clips—will be removed. Annotations will be verified for consistency. Metadata from the text file will be reviewed using Pandas to ensure compatibility with the compiled video, though timestamps will not be used due to the nature of the video compilation.

3.4.4 Data Analysis

The YOLOv8 model will be trained on the custom dataset to detect objects and track movement. Using the Ultralytics library, the model will be fine-tuned with specific parameters including the number of training epochs, image size, and enabled training progress plots. Training will be based on the data.yaml file and set to detection mode. Post-detection, a centroid-based tracking algorithm will be used to track object movement across frames. Displacement will be calculated using Euclidean distance, and speeds will be estimated with a calibrated scaling factor. OpenCV will handle the frame-by-frame analysis in real time. The model will output bounding boxes, class probabilities, and speed estimates, which will be saved as annotated video frames and CSV files for further evaluation.

3.4.5 Data Deployment

The system will be deployed as a real-time RTA monitoring application using Streamlit. The trained YOLOv8 model will be integrated into a Streamlit app to process uploaded videos. The user interface will include a title page, a section for uploading MP4 files from X, real-time object detection and speed tracking, and a download link for the output video. Videos will be processed frame-by-frame, with YOLOv8 detecting objects and the tracking algorithm computing velocities. Annotated frames will be compiled into a video using FFmpeg. The app will be tested locally and deployed on a cloud platform (e.g., Heroku) for access by stakeholders. The final output will be a web-based interface displaying annotated videos with bounding boxes, class labels, and speed estimates, supporting NTSA and law enforcement in monitoring traffic incidents.

3.4.6 Data Validation and Evaluation

The system's performance will be evaluated using quantitative metrics and stakeholder feedback. The YOLOv8 model will be assessed on a 20% validation set using key metrics: precision and recall (targets > 0.90), mAP50 (target > 0.95), and mAP50-95 (target > 0.65). Speed tracking accuracy will be validated against ground truth data (e.g., speedometer readings), with a target Mean Absolute Error (MAE) below 5 km/h. The Streamlit app will be evaluated for real-time performance (target FPS > 20), and user feedback will be collected from NTSA representatives. To test robustness, the system will be run on diverse video conditions (night, rain, rural roads), aiming for a detection accuracy above 85% across all scenarios. The dataset split will ensure unbiased evaluation, and all metrics will be computed using Python libraries such as Ultralytics and scikit-learn, with results visualized through confusion matrices and precision-recall curves.

3.5 Ethical Considerations

The use of X video footage and deployment of a real-time monitoring system raises several ethical considerations:

3.5.1 Institutional Approval and Ethical Oversight

This study will be submitted to the USIU-A Ethics Review Committee for formal approval to ensure compliance with institutional ethical guidelines and Kenya's Data Protection Act (2019).

The approval process will confirm that data collection, processing, and storage methods align with national and academic research standards. Any future publications or public deployments of the system will be guided by this review to ensure transparency and accountability.

3.5.2 Data Privacy

Although the videos are publicly available on X, the study ensures that no personally identifiable information (PII), such as faces, license plates, or usernames, is stored, displayed, or shared. Video files are securely stored on local encrypted drives with restricted access. All raw video data will be permanently deleted after analysis is complete (by June 2025), and only anonymized outputs (e.g., object classes and coordinates) will be retained for model evaluation.

3.5.3 Participant Consent

As the data is collected from publicly posted videos, obtaining individual consent is not feasible. However, the study adheres to X's content usage policies and terms of service. In addition, a public statement about the study will be posted on X using hashtags like #StaySafe, inviting users to learn about the research and opt out by contacting the research team if their content is identifiable. This ensures a minimal degree of informed awareness and respect for public contributors.

3.5.4 Potential Impact on Stakeholders

The system is intended to support public safety initiatives by enabling NTSA, law enforcement, and emergency responders to monitor road conditions and RTAs in real time. However, there is a risk of misuse or over-reliance on automated alerts, especially if false positives or model bias occurs. To mitigate this, the Streamlit interface includes clear confidence scores and disclaimers, and all outputs are intended to be used as decision-support tools—not as replacements for official reports or legal evidence.

Stakeholders will be provided with training materials and onboarding sessions to ensure responsible use and interpretation of the system's outputs. Feedback loops will also be built in, allowing authorized users to flag inaccuracies, which may help improve the model iteratively.

3.6 Feasibility Considerations

3.6.1 X Feasibility of Deployment in the Kenyan Context

Given Kenya's limited access to high-performance computing infrastructure and stable internet connectivity in some areas, the proposed system is designed with resource-efficiency in mind. The object detection and speed tracking components will be tested and optimized on consumer-grade hardware (e.g., standard laptops or desktop machines with mid-range GPUs) to ensure feasibility in real-world deployments.

Furthermore, cloud-based alternatives such as deployment via Streamlit Cloud will be explored to enable scalable access across urban centers. These platforms offer accessible solutions for stakeholders such as traffic authorities who may not have the capacity to maintain dedicated local servers.

To mitigate bandwidth limitations, the system is designed to process pre-recorded video clips locally before uploading lightweight summaries (such as alert flags, timestamps, and object metadata) to cloud dashboards. This reduces reliance on continuous streaming and makes the system more adaptable to constrained environments.

In future implementations, compressed models (e.g., YOLOv8n or YOLOv8s) and quantized versions will be considered to reduce memory usage and improve inference time without significantly sacrificing accuracy.

3.7 Chapter Summary

Chapter 3 presents the comprehensive methodology adopted for developing a real-time road traffic accident (RTA) detection and speed tracking system tailored to Kenya's traffic conditions. The approach integrates computer vision and machine learning, particularly the YOLO model, for detecting objects such as vehicles, pedestrians, motorcycles, and accidents from video footage sourced from the X platform (formerly Twitter).

The chapter begins by outlining a quantitative experimental research design structured around two main components: a real-world case study based in Kenya and a prototype system built

using Streamlit. This system processes crowd-sourced videos to detect RTAs and estimate vehicle speeds in real-time, contributing to data-driven road safety management.

Detailed experiment methodology follows, highlighting model development, including fine-tuning YOLO model and implementing a centroid-based speed tracking algorithm. The step-by-step design procedure covers data acquisition from social media, frame extraction, annotation, cleaning, training, analysis, and real-time deployment. Data is rigorously validated using metrics such as mAP, precision, recall, and MAE to ensure model accuracy and robustness across diverse conditions.

Finally, the chapter addresses ethical considerations, focusing on data privacy, consent, and the responsible deployment of surveillance technologies. Measures are taken to protect identities and inform the public, while safeguards against misinterpretation and over-reliance on automation are implemented to ensure responsible system use by stakeholders like NTSA.

Through a structured and ethically sound methodology, this chapter lays the foundation for building a practical and scalable system that leverages AI to improve traffic incident detection and promote road safety in Kenya.

CHAPTER 4.0: DESIGN/EXPERIMENT RESULTS

4.1 Introduction

This chapter presents the results from building and testing the system. The project focuses on addressing road accidents and speeding in Kenya by using video footage to detect accidents, identify road users, and estimate vehicle speeds. It outlines the process of dataset creation, model training, object detection results, and speed estimation results.

4.2 Presentation of Artifacts/Results

4.2.1 Dataset Development

To support the object detection task, a custom dataset was created using real-world traffic footage from Kenya. This dataset enabled the training of two object detection models based on different class structures:

- **Model A** was trained on a 5-class dataset: *Accident*, *Passenger*, *Motorcycle*, *Pedestrian*, *Car*.
- **Model B** focused on 4 classes, excluding *Passenger*, to emphasize relevant traffic scenarios.

Both datasets were annotated in YOLO format, and sample annotated images with bounding boxes were generated (see Figure 2). The datasets were divided into training and validation subsets to support reliable model evaluation.



Figure 2: Sample of annotated image with bounding boxes (4 classes)

This dual-dataset strategy supported the flexibility to evaluate detection under different labeling schemes and ultimately enhanced robustness when combining outputs.

4.2.2 Object Detection Models

Each model produced detection outputs in the form of bounding boxes, class labels, and confidence scores. Their predictions were visually distinct, reflecting the different focus of each model. **Figure 3** shows a comparison between the outputs of Model A and Model B when applied to the same video frame.

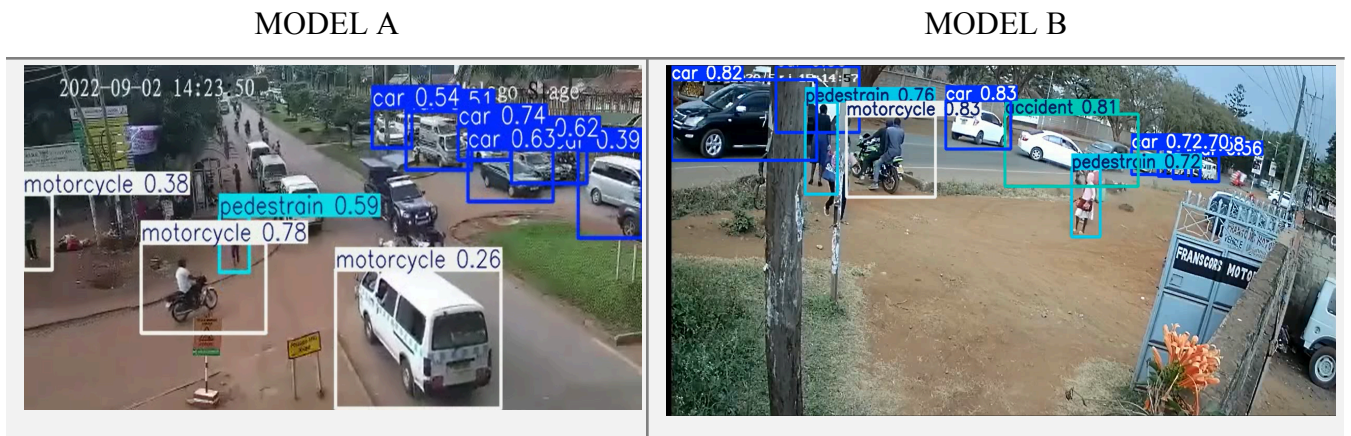


Figure 3: Comparison between Model A and Model B detection outputs

A detailed performance analysis, including precision, recall, and mean Average Precision (mAP), is provided in **Section 4.3**.

4.2.3 Combined Detection System

To leverage the strengths of both models, a combined detection system was developed. The integration was guided by logic that prioritized detection accuracy and minimized redundancy. Key features included:

- **Non-Maximum Suppression (NMS):** To avoid duplicate detections from overlapping bounding boxes.
- **Confidence Thresholding:** To discard low-certainty predictions.

The resulting system generated annotated video frames showing object class labels and their confidence levels, improving detection in crowded and complex urban scenes.



Figure 4: Combined Object Detection model output

4.2.4 Speed Estimation

The system incorporated a vehicle speed estimation module, enabling real-time assessment of vehicle motion. By tracking the displacement of vehicles (identified through bounding boxes) across consecutive frames, speeds were calculated and overlaid on the video output.

- Vehicles were uniquely identified using object tracking techniques.
- Speeds were estimated and displayed dynamically, ranging typically from 0–100 km/h — suitable for urban traffic scenarios.

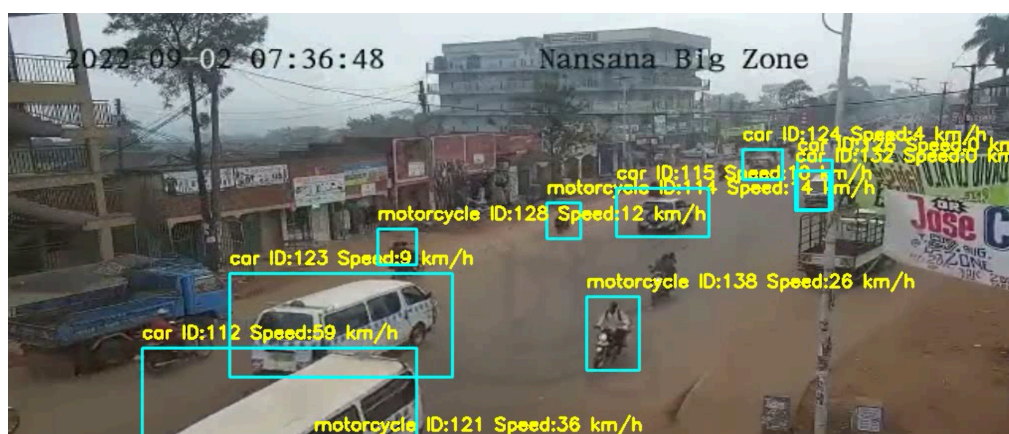


Figure 5: Speed Estimation model output

4.3 Analysis of Result

4.3.1 Model A – 5-Class Performance

<i>Class</i>	<i>Images</i>	<i>Instances</i>	<i>Precision</i>	<i>Recall</i>	<i>mAP50</i>	<i>mAP50-95</i>
<i>ALL</i>	100	1160	0.931	0.896	0.963	0.689
<i>Accident</i>	85	90	0.986	0.978	0.991	0.830
<i>Passenger</i>	29	44	0.901	0.841	0.923	0.622
<i>Motorcycle</i>	70	126	0.936	0.812	0.958	0.709
<i>Pedestrian</i>	91	242	0.926	0.936	0.973	0.614
<i>Car</i>	90	658	0.906	0.911	0.970	0.672

Model A was evaluated on test data that included scenes with multiple traffic participants (cars, motorcycles, pedestrians, passengers) as well as accident instances. It achieved high performance in detecting **cars** and **motorcycles**, which were its dominant training categories. However, its accuracy for detecting accidents and pedestrians was slightly lower, reflecting a broader but less focused learning scope.

4.3.2 Model A – 4-Class Performance

<i>Class</i>	<i>Images</i>	<i>Instances</i>	<i>Precision</i>	<i>Recall</i>	<i>mAP50</i>	<i>mAP50-95</i>
<i>ALL</i>	100	1160	0.931	0.896	0.963	0.689
<i>Accident</i>	85	90	0.986	0.978	0.991	0.830
<i>Passenger</i>	29	44	0.901	0.841	0.923	0.622
<i>Motorcycle</i>	70	126	0.936	0.812	0.958	0.709
<i>Pedestrian</i>	91	242	0.926	0.936	0.973	0.614
<i>Car</i>	90	658	0.906	0.911	0.970	0.672

Model B focused specifically on accidents involving **cars**, excluding less critical classes such as *Passenger*. As a result, it demonstrated improved performance in scenarios involving car collisions. It achieved higher recall for **Accident** and **Car** categories compared to Model A, benefiting from its narrower and more specialized focus.

4.3.3 Combined Model Detection Performance

The combined model was tested on multiple video sequences featuring various traffic scenes, including normal flow, congestion, and accidents. The key observations from the visual outputs include:

- **Object Detection Consistency:** The system accurately identified and labeled key road users such as cars, motorcycles, pedestrians, and passengers across different frames. Bounding boxes remained stable and correctly tracked the objects throughout motion sequences.
- **Accident Recognition:** Accident scenarios were correctly flagged, with the system displaying the “accident” label only when there was clear evidence of a collision or road incident in the footage. This confirms that the accident detection logic responds appropriately based on visual cues in the frame.
- **Multi-class Clarity:** Each class label was distinguishable in real-time, and the system could detect multiple road users simultaneously without overlapping or misclassification in most cases.

4.3.4 Speed Estimation Performance

Speed estimation was integrated into the detection output as an overlay on the bounding boxes. The following points summarize the qualitative evaluation:

- **Accuracy Relative to Motion:** Vehicles moving at visibly higher speeds had correspondingly higher speed values displayed, while stationary or slow-moving vehicles showed lower or near-zero speed estimates.
- **Frame Consistency:** Speed readings updated in real-time with each frame, and fluctuations were smooth — indicating reliable tracking without major spikes or drops.
- **Multiple Vehicle Handling:** In scenarios with several moving vehicles, the system successfully maintained individual speed estimates for each, and correctly matched speeds with the associated object.

Although no ground truth speed values were used, the visual inspection demonstrated that the estimates were reasonable and aligned with observed motion behavior in the video.

4.4 Discussion of Findings

The results obtained from the system implementation address the core research objectives effectively. Real-time object detection using YOLOv8 proved capable of identifying road users and flagging potential accidents with visual accuracy. The models, particularly the combined one, demonstrated stability in detecting various classes such as cars, pedestrians, motorcycles, and accident events across different scenes.

Speed estimation, though evaluated qualitatively, showed coherent alignment with observed vehicle motion, providing real-time updates with frame consistency. This indicates the feasibility of integrating object detection with speed tracking in a unified pipeline. The findings suggest that such a system can enhance situational awareness for urban traffic monitoring and serve as a basis for future real-time alert mechanisms. Additionally, the ability to visually track speed and accident events can support both immediate interventions and inform long-term strategies for road safety management in Kenya.

4.5 Chapter Summary

This chapter presented the experimental results and system artifacts developed for real-time traffic accident detection and speed estimation using video footage. A custom dataset was created and used to train two YOLOv8-based object detection models, with one focused on broader classification and the other tailored for accident-specific scenarios.

Both models were evaluated individually and later integrated into a combined system for enhanced detection accuracy. Speed estimation was achieved by tracking object movement across frames, and visual inspection confirmed the consistency of the results. Performance metrics such as precision, recall, and mAP indicated strong model reliability. Overall, the system demonstrated its capability to detect road users, recognize accident events, and estimate vehicle speeds in real-time, addressing the research objectives effectively.

CHAPTER 5.0: EVALUATION, VALIDATION AND DEPLOYMENT

5.1 Introduction

This chapter discusses how the system was evaluated and validated after development. It outlines the criteria used to measure performance, the methods used to test and verify the results, and how the system was assessed in terms of accuracy, reliability, and usefulness. The final part covers the deployment of the model and how it can be used in real-world scenarios.

5.2 Evaluation Criteria

The system was evaluated using both quantitative metrics for object detection and qualitative assessment for speed estimation. The chosen criteria are:

5.2.1 Object Detection Evaluation Metrics

- **Precision:** Measures the proportion of correct positive predictions out of all positive predictions made.
- **Recall:** Measures the proportion of correct positive predictions out of all actual positives.
- **F1 Score:** Harmonic mean of precision and recall; balances false positives and false negatives.
- **mAP@0.5 (Mean Average Precision at IoU = 0.5):** Indicates overall object detection performance at a 0.5 IoU threshold.
- **mAP@0.5:0.95:** Measures detection performance over multiple IoU thresholds (0.5 to 0.95), offering a more comprehensive view.
- **Confusion Matrix:** Shows the count of correct and incorrect predictions for each class.
- **Precision-Recall Curves:** Visualize the trade-off between precision and recall for each object class.

5.2.2 Speed Estimation Evaluation Criteria

- **Consistency Across Frames:** Checks whether estimated speeds remain stable for the same object across consecutive frames.

- **Multi-object Tracking Performance:** Evaluates the system's ability to estimate speeds for multiple moving objects simultaneously.
- **Visual Accuracy:** Assesses if the speed overlays appear logically consistent with actual motion in the video (based on visual inspection).
- **Qualitative Validation:** Since no ground truth speed data was available, results were validated through careful visual review of outputs.

These criteria were selected to provide a balanced evaluation of the model's performance in real-world video analysis, covering both detection accuracy and usability in traffic monitoring contexts.

5.3 Validation Process

The two models, A and B were validated on separate annotated test datasets not seen during training. Additionally, the ensemble model was evaluated to check improvements in detection through merged outputs using Non-Maximum Suppression (NMS).

5.3.1 Object Detection Validation

Steps followed:

- Ran inference on validation datasets using Model A and Model B.
- Compared predicted bounding boxes and class labels to ground truth annotations.
- Computed evaluation metrics using YOLOv8's internal module and visualized results with matplotlib and seaborn.

Visual tools used:

- Precision-Recall curves
- Confusion matrices
- F1-score bar plots

The ensemble model was tested on the same datasets, and results showed improved detection accuracy and reduced false positives after applying NMS.

5.3.2 Speed Estimation Validation

Speed estimation was performed using the ensemble model. No dedicated ground truth dataset was available for speed validation, so the process was observational. Test video clips representing varied traffic conditions (heavy, light, and static) were used to visually inspect the accuracy and consistency of speed predictions across frames.

Key validation checks:

- Visual consistency of object speeds across consecutive frames
- Realistic variation in speed depending on object motion
- Performance in scenes with multiple moving and static objects

This process confirmed the system's practical usability under different traffic conditions, even though it was qualitative in nature due to the absence of exact ground truth speed labels.

5.4 Validation Results

5.4.1 Model A – 5-Class

Model A was validated on a diverse dataset with five object classes. The results confirmed that the model performs well across all classes, especially in identifying accidents and pedestrians.

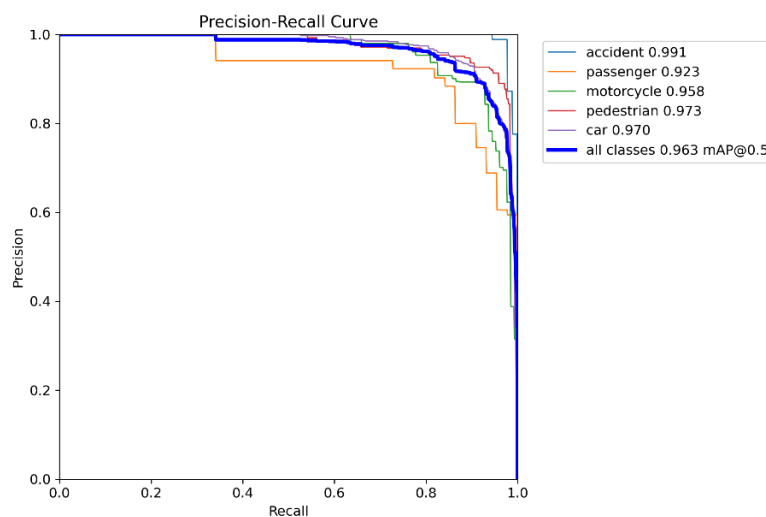


Figure 5.1: Precision-Recall Curve for Model A

The Precision-Recall (PR) curve for Model A shows strong classification capability. The accident class curve is closest to the top-right corner, indicating exceptional detection accuracy. Other classes such as motorcycle, car, and pedestrian also demonstrated high average precision values, though the passenger class had a relatively lower curve, reflecting its smaller sample size and visual ambiguity.

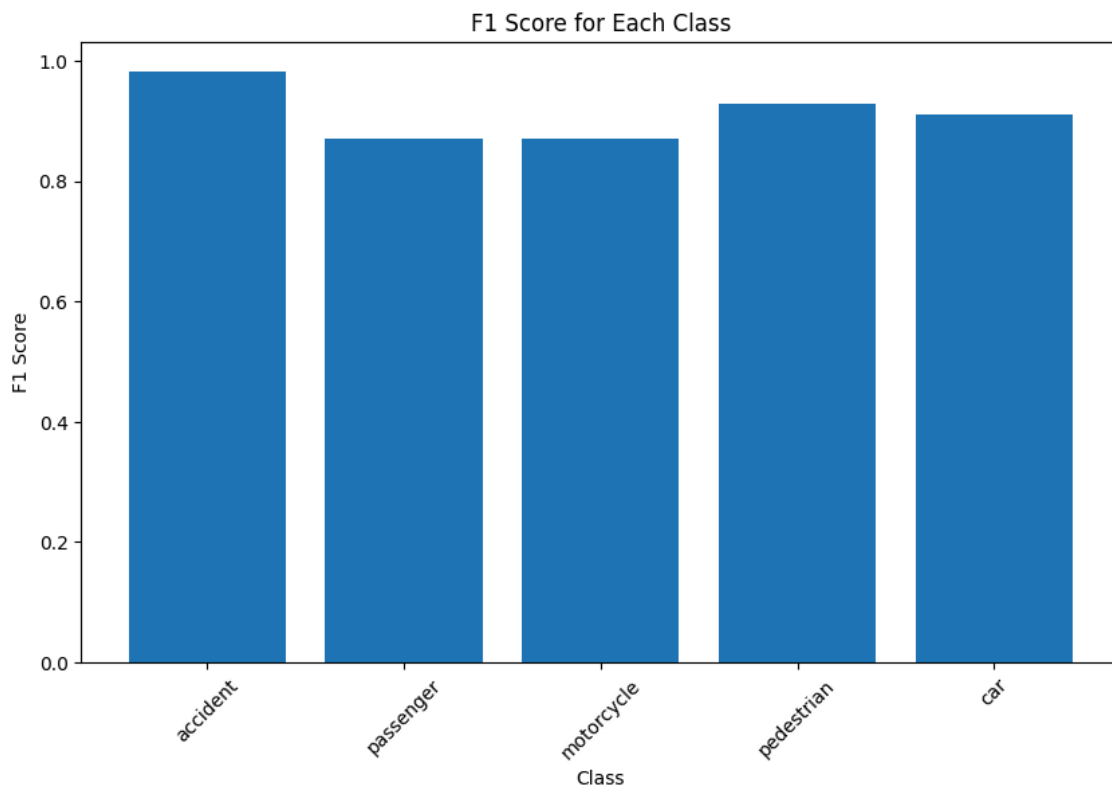


Figure 5.2: F1 Score per Class – Model A

The F1 score bar chart confirms this trend, with the highest score observed in the accident class (≈ 0.98), followed closely by pedestrian and motorcycle. The passenger class showed the lowest F1 score (~ 0.87), consistent with its PR curve.

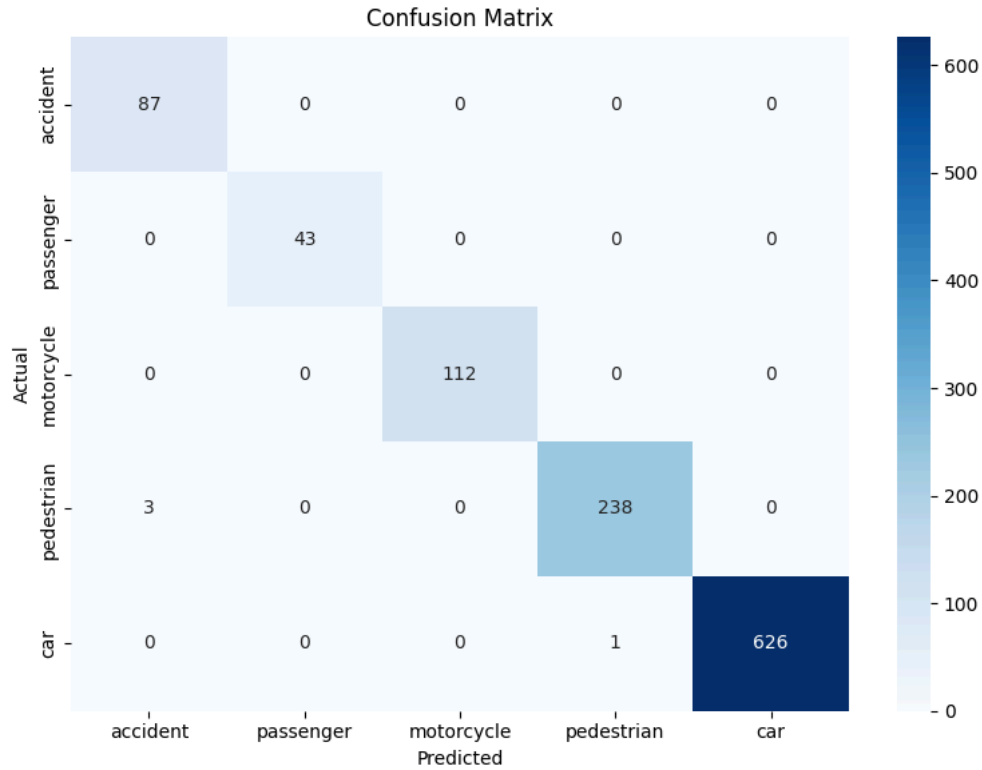


Figure 5.3: Confusion Matrix – Model A

The confusion matrix provides further insight into model reliability. Diagonal dominance indicates that most predictions were accurate. Only minor misclassifications were observed—such as a few pedestrians being labeled as accidents and one car being mislabeled as a pedestrian—indicating a high level of class separation and model precision.

5.4.2 Model B – 4-Class

Model B was designed with a narrower focus, which improved performance, particularly for high-priority categories like accidents.

- The PR curve for Model B (Figure 5.4) shows that all classes maintain precision and recall values near 1.0, with the accident class again dominating the performance spectrum. This consistency indicates very low false positive and false negative rates across all categories.

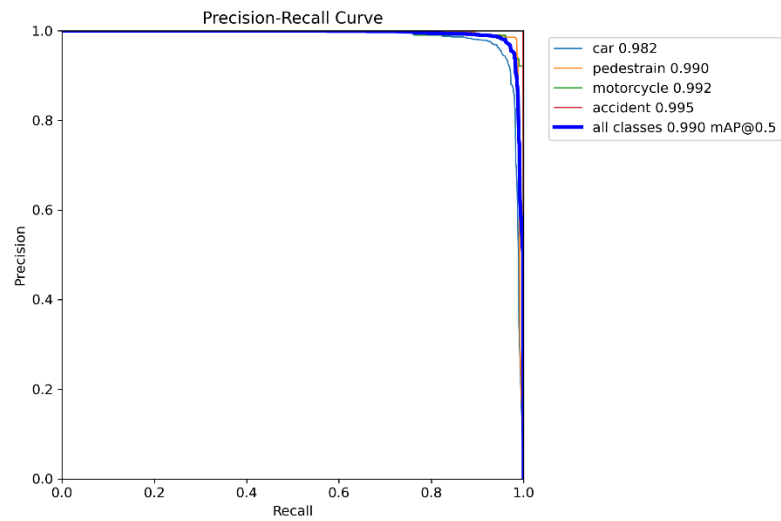


Figure 5.4: Precision-Recall Curve for Model B

The F1 scores reinforce this, with every class scoring between ~ 0.95 and ~ 0.99 . The accident and pedestrian classes achieved the highest values, confirming that the model performs extremely well in safety-critical contexts.

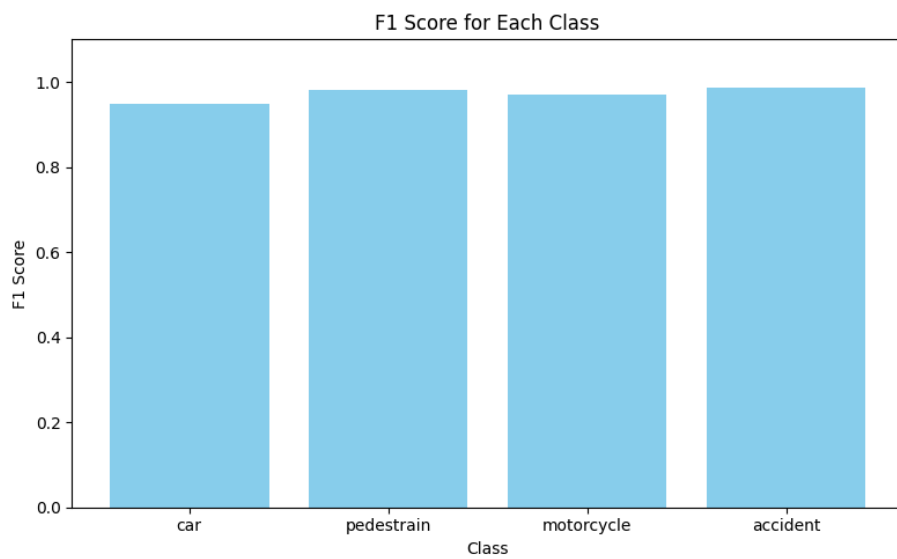


Figure 5.5: F1 Score per Class – Model B

- The confusion matrix is nearly perfect, with all objects correctly classified except for a single instance where a motorcycle was mislabeled as a pedestrian. No other misclassifications were recorded, validating the model's robustness.

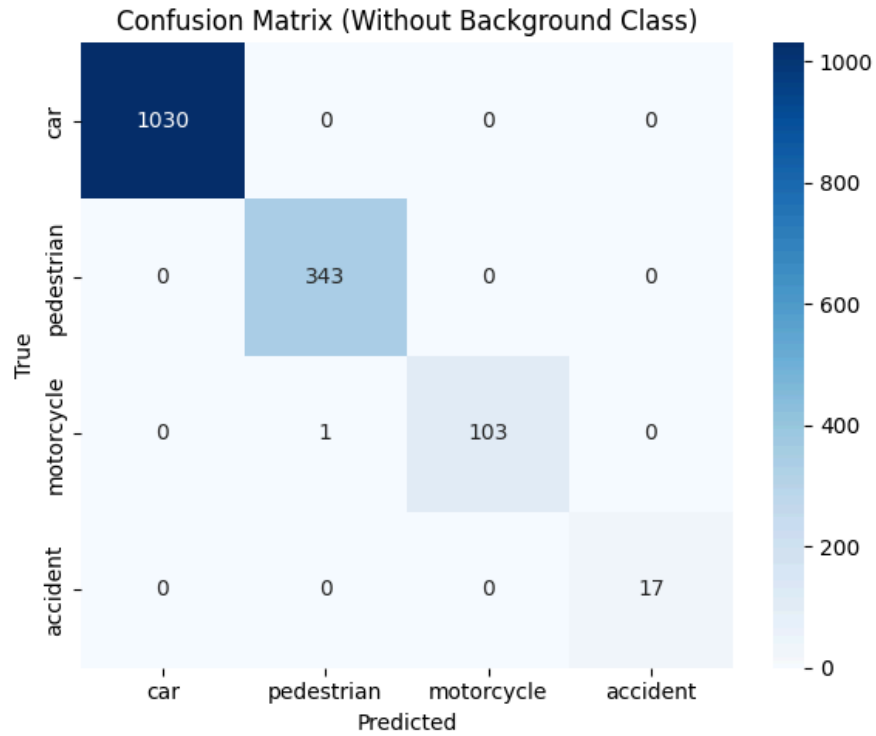


Figure 5.6: Confusion Matrix – Model B.

5.4.3 Speed Estimation Validation

The speed estimation module was qualitatively evaluated due to the absence of ground truth speed data. Nevertheless, the visual inspection of overlaid speed values across various test videos confirms the reliability of the implementation.

5.5 Discussion of Validation

The validation results provide strong evidence of the effectiveness and generalizability of the developed models. The combination of Model A and Model B through an ensemble approach proved particularly beneficial in mitigating the limitations seen when each model was used independently.

Key observations from the validation include:

- **Strong Performance in Real-World Conditions:** The models demonstrated high accuracy and robustness when tested on previously unseen data, suggesting good generalization capabilities. This is essential for practical deployment where traffic scenes vary widely in lighting, weather, and density.
- **Effectiveness of the Ensemble Approach:** The ensemble model, which combined predictions using Non-Maximum Suppression (NMS), improved the overall detection reliability. It reduced false positives and corrected several misclassifications observed in the individual models.
- **High Accuracy for Key Classes:** The system performed best in detecting cars and accidents, which are the most critical for road safety applications. These results align well with the core objective of the project: to assist in crash detection and improve traffic monitoring accuracy.
- **Room for Improvement in Minority Classes:** The lower mAP50-95 scores for passengers and pedestrians indicate areas where performance could be improved. These results likely stem from class imbalance in the training data or insufficient feature diversity for these classes. Additional annotated examples or data augmentation could help resolve this issue in future work.
- **Practical Readiness:** Overall, the validation process confirms that the models, especially the ensemble version, are suitable for real-world deployment. They offer robust detection capabilities, reasonable inference speed, and reliable performance under varying conditions.

These findings validate the technical soundness of the model design choices and emphasize the importance of leveraging ensemble strategies in object detection tasks, particularly in dynamic environments such as road surveillance systems.

5.6 Deployment

The developed Traffic Intelligence App was deployed using Streamlit Cloud to enable real-time accessibility and interaction. The deployment process began by organizing all project components into a well-structured directory. This included placing trained YOLO detection

models in a dedicated models folder, processed videos in a videos folder, and supporting scripts such as `video_detect.py`, `speed_estimation.py`, and the main interface script `streamlit_app.py` in the root directory. A `requirements.txt` file was generated to capture all necessary Python dependencies, including packages such as `streamlit`, `opencv-python`, `ultralytics`, `numpy`, `Pillow`, and `deep_sort_realtime`.

After setting up the project locally, version control was initialized using Git, and the entire repository was pushed to GitHub under the repository name `Graduation-project`. Following this, the deployment was completed via Streamlit Cloud. The GitHub repository was linked to Streamlit, and the main script `streamlit_app.py` was specified as the entry point. Upon successful configuration and deployment, Streamlit Cloud provided a public URL through which users can interact with the Traffic Intelligence App in real time. This deployment approach ensures scalability, ease of access, and effective demonstration of the app's real-world applicability in detecting road incidents and estimating vehicle speed.

5.7 Chapter Summary

This chapter covered the evaluation, validation, and deployment of the Traffic Intelligence App. The models were assessed using precision, recall, F1 score, mAP, and visual tools like confusion matrices. An ensemble approach combining two YOLOv8 models using Non-Maximum Suppression improved detection accuracy and reduced false positives.

Speed estimation was validated visually across varied traffic scenes, showing consistent performance despite the absence of ground truth speed data. The models performed best in detecting cars and accidents, aligning with the project's safety goals, while lower performance in passenger and pedestrian detection highlighted the need for more balanced data.

Deployment was successfully done via Streamlit Cloud, enabling public access and real-time monitoring in practical settings.

CHAPTER 6.0: DISCUSSION, CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

This chapter presents a comprehensive discussion of the study's findings, followed by conclusions and recommendations based on the results. The section revisits the research objectives and offers insights into how they were addressed. Additionally, it outlines the limitations encountered, suggests practical and theoretical implications, and proposes areas for further research in the field of data-driven traffic monitoring.

6.2 Research Summary

The study aimed to develop and evaluate a real-time Traffic Intelligence App for accident detection and vehicle speed estimation using video footage. The project was motivated by the high rates of road accidents in Kenya and the need for intelligent traffic monitoring tools. The key objectives were:

- To build a system that can detect road users and traffic incidents using object detection models.
- To estimate the speed of vehicles in real time using video analysis.
- To deploy the system in a manner accessible for real-world use.

The methodology involved collecting and annotating a custom dataset from Kenyan traffic scenes, training two YOLOv8-based models (Model A with 5 classes and Model B with 4 classes), and developing an ensemble detection system. The models were evaluated using precision, recall, F1 score, mAP, and visual analysis tools. Speed estimation was validated through visual inspection due to the lack of labeled speed data.

The significance of the research lies in its potential application for real-time road safety monitoring, traffic management, and post-incident analysis.

6.3 Results Interpretation and Discussion

The results demonstrated that both object detection models performed well, with particularly high accuracy in detecting cars and accidents. Model A offered broader class coverage, while Model B excelled in accident-related scenarios. Combining both models using Non-Maximum Suppression led to improved detection accuracy and fewer false positives.

The speed estimation module, though evaluated qualitatively, produced consistent and realistic outputs across a range of traffic videos. Vehicles moving quickly were assigned higher speed values, while static vehicles showed near-zero speeds.

These findings align with existing literature that supports the effectiveness of YOLO-based models in real-time applications. The ensemble approach, in particular, confirms research suggesting that combining models can enhance robustness and performance.

The study successfully met its objectives and demonstrated that a lightweight, efficient, and deployable traffic monitoring system can be built using modern computer vision techniques.

6.4 Limitations

Despite the promising results, the study faced several limitations:

- **Lack of Ground Truth Speed Data:** Speed estimation was validated visually, which limits quantitative performance evaluation.
- **Environmental Diversity:** The dataset did not include diverse weather or lighting conditions, which may affect model generalizability in real-world deployment.
- **Hardware Limitations:** Deployment was tested on cloud platforms but not on embedded or edge devices, which may have different performance constraints.

6.5 Conclusions

This study successfully developed a functional and deployable Traffic Intelligence App capable of detecting road users, identifying traffic incidents, and estimating vehicle speeds in real time.

The combination of two YOLOv8 models into an ensemble system enhanced detection accuracy, particularly for accidents and cars—key classes for road safety.

Although speed estimation was qualitatively validated, the module showed practical promise for real-time applications. Overall, the system provides a scalable and effective solution for urban traffic monitoring, with potential use in road safety enforcement, accident detection, and transport planning.

6.6 Recommendations

Based on the findings, the following recommendations are proposed:

Practical Recommendations

- **Integration with Urban Surveillance Systems:** The app can be integrated into citywide CCTV networks to assist in real-time traffic incident monitoring.
- **Use in Law Enforcement:** The system could be further developed to trigger alerts for potential overspeeding or crash incidents to aid traffic authorities.
- **Edge Deployment:** Optimize the system for edge devices (e.g., Raspberry Pi, NVIDIA Jetson) to reduce reliance on cloud infrastructure.

Theoretical/Technical Recommendations

- **Speed Validation Dataset:** Develop or acquire labeled datasets with ground truth vehicle speeds to enable quantitative evaluation of speed estimation accuracy.
- **Multimodal Fusion:** Explore the use of additional inputs (e.g., audio, sensor data) to enhance accuracy in complex environments.

6.7 Recommendations for Further Research

- **Longitudinal Field Testing:** Deploy the system for extended periods in live traffic environments to assess long-term reliability and adaptability.

- **Behavioral Analysis:** Extend the system to analyze driver or pedestrian behavior, which could assist in predicting potential accidents before they occur.
- **Traffic Flow Prediction:** Incorporate AI models to forecast traffic congestion or flow based on detected patterns.
- **Comparative Model Analysis:** Evaluate other real-time models (e.g., YOLO-NAS, EfficientDet) against YOLOv8 to determine optimal architecture for traffic intelligence tasks.

6.8 Chapter Summary

This chapter discussed the findings in light of the research objectives and existing literature. The system achieved strong performance in detecting key road users and traffic incidents and showed potential for practical deployment. While limitations exist, particularly in speed estimation validation and dataset diversity, the project contributes meaningfully to the field of intelligent traffic monitoring. Conclusions and recommendations were drawn, along with suggestions for future research directions to further enhance system performance and applicability.

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APPENDICES

A. Supplementary Materials

1. Work Plan

- Weekly breakdown of project activities (see timeline below).

2. Consent Statement

- Data sourced publicly via Twitter API; anonymized per ethical protocols.

3. Data Collection Tools

- Python (OpenCV for video frame extraction, Ultralytics for model training,)
- LabelImg for manual annotation

4. Experimental Protocols

- Video preprocessing: Frame extraction, resizing, and quality filtering.
- Annotation: Manual labeling of frames for Car, Pedestrian, Motorcycle, and Accident classes.
- YOLOv8 training: Fine-tuning with Ultralytics library
- Speed tracking: Centroid-based tracking using YOLOv8 bounding box coordinates; speed calculated.

5. Sample Dataset Fields

- Video_ID: Unique identifier for X video.
- Frame_Number: Sequential frame ID in composite video.
- Object_Label: Class (Car, Pedestrian, Motorcycle, Accident).
- Bounding_Box: Coordinates (x, y, width, height).
- Speed_Estimate: Calculated speed (km/h) for detected objects.

6. Algorithms Used

- YOLOv8: Single-stage object detection for real-time classification and localization.
- Centroid Tracking: For tracking object movement across frames.
- OpenCV: For frame extraction and preprocessing.

7. Additional Analyses

- Mean Average Precision (mAP50, mAP50-95): For evaluating YOLOv8 detection accuracy.
- Mean Absolute Error (MAE): For validating speed estimates against ground truth (e.g., speedometer readings).
- Confusion matrices and precision-recall curves: For model performance visualization.
- Robustness testing: Evaluating detection accuracy across diverse conditions (night, rain, rural roads).

8. Project Code

The following is a link to my google colab

https://colab.research.google.com/drive/15wi_-qE5MFaSgrT7422rXcccbhROqnF#scrollTo=ub2SJcPIE-QM

9. Deployment

The following is a link to my project deployment

<https://graduation-project-nfufoxa9rwcifed9fzirr.streamlit.app/>

APPENDIX I: INTRODUCTION LETTER

I hope this letter finds you well. My name is Neema Ndanu Musomba, a Data Science and Analytics student at USIU-Africa. I am currently undertaking a research project titled: “Real-Time Traffic Detection and Speed Estimation in Kenya Using YOLOv8.”

I am writing to kindly request your assistance in granting academic permission to access and use publicly available, non-sensitive video content from the X platform (formerly Twitter), and in supporting the ethical review process for this research project. Your support will be invaluable in shaping the direction and impact of this study.

The purpose of this research is to explore how computer vision and artificial intelligence, specifically the YOLOv8 object detection model, can be applied to detect road traffic accidents and estimate vehicle speeds in real time.

To conduct this research, I will use publicly available traffic-related videos from the X platform, which regularly features real-time road incidents. These videos will be used to train and evaluate an AI model for detecting vehicles, motorcycles, and pedestrians, and for estimating their movement and speed. No direct human data will be collected, and all personally identifiable information will be excluded or anonymized in accordance with the Kenya Data Protection Act (2019).

Your support in this process will be greatly appreciated. All data will be used strictly for academic purposes, and the findings will be shared with you upon request.

Should you require any additional information, please feel free to contact me at nmusomba@usiu.ac.ke or +254 742 502 862. Thank you in advance for your time and consideration. I look forward to your positive response.

Yours sincerely,

Neema Ndanu Musomba

APPENDIX II: CONSENT FORM

Title of the Study: Real-Time Traffic Detection and Speed Estimation in Kenya Using YOLOv8

Investigator: Neema Ndanu Musomba

Faculty Advisor: Dr. Gabriel Okello, School of Science and Technology, USIU-Africa

You are invited to participate in a research study on Real-Time Traffic Detection and Speed Estimation in Kenya Using YOLOv8. The purpose of the study is to explore how computer vision and artificial intelligence, specifically the YOLOv8 object detection model, can be applied to detect road traffic accidents and estimate vehicle speeds in real time. The study will be conducted by Neema Ndanu Musomba, a student of School of Science and Technology, USIU-Africa, under the guidance of Dr. Gabriel Okello, faculty advisor.

Participation may include a brief consultation or review of system implementation strategies and may take approximately 20–30 minutes, depending on the level of feedback you choose to provide. Notes may be taken with your permission to support academic documentation.

There are no anticipated risks associated with your participation. While there are no direct personal benefits, your input will contribute to the development of safer, data-driven road monitoring tools that may support agencies such as NTSA or emergency responders.

All information collected during the study will be kept confidential and anonymous. No individual or organization will be identified in the research report or any publication. The data will be stored in a secure location and only accessible by the research team.

Your participation in this study is voluntary. You have the right to refuse to participate, skip specific feedback areas, or to withdraw from the study at any time without penalty or loss of benefits.

If you have any questions or concerns about the study, please contact the investigator at nmusomba@usiu.ac.ke or the faculty advisor at gokello@usiu.ac.ke. If you have questions about

your rights as a research participant, you may contact the Institutional Review Board at USIU-Africa at irb@usiu.ac.ke.

I have read and understand the information provided in this consent form. I voluntarily agree to participate in this study and understand that I can withdraw at any time without penalty or loss of benefits.

Signature: _____ Date: _____

APPENDIX III: DEBRIEF FORM

UNITED STATES INTERNATIONAL UNIVERSITY - AFRICA
DEBRIEFING FORM/SCRIPT
REAL-TIME TRAFFIC DETECTION AND SPEED ESTIMATION IN KENYA USING
YOLOv8

Thank you for participating in this research study conducted by Neema Ndanu Musomba from School of Science and Technology, USIU-Africa.

This study aims to develop a real-time traffic monitoring system using the YOLOv8 object detection algorithm to automatically detect road traffic accidents and estimate vehicle speeds. The system is trained using publicly available traffic-related videos from the X platform (formerly Twitter), which were processed and manually annotated to label objects such as vehicles, motorcycles, pedestrians, and accident events. No personal or sensitive information is collected.

You have the right to withdraw your data collected from or about you prior to debriefing, without penalty or loss of benefits to which you are otherwise entitled. Please initial below if you do, or do not, give permission to have your data included in the study:

_____I give permission for the data collected from or about me/ institution to be included in the study.

_____I DO NOT give permission for the data collected from or about me/ institution to be included in the study.

The researcher conducting this study is Neema Ndanu. Please ask any questions you have now. If you have questions later, you may contact the researcher at [nmusomba@usiu.ac.ke] or at +254 742 502 862.]. If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the USIU-Africa Institutional Review Board (IRB) Chairperson at irb@usiu.ac.ke.

Your signature below indicates that you have been debriefed, and have had all of your questions answered.

Name of Researcher

Signature

Date

Name of Participant

Signature

Date

APPENDIX IV: DATA COLLECTION

This research study did not involve human respondents, surveys, or interviews.

Data for this study was collected exclusively from publicly available road traffic videos shared on the X platform (formerly Twitter) between 2023 and 2025. Videos were downloaded from traffic-related accounts such as @Ma3Route using relevant keywords (e.g., "*accident*", "*crash*", "*traffic*"). The videos were compiled into a dataset and used for object detection and speed estimation modeling.

No personally identifiable information was collected or used in this study. All data was handled in compliance with the Kenya Data Protection Act (2019) and used solely for academic research purposes.